

Differential Evolution with Local Search Algorithms for Data Clustering: A Comparative Study



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Abstract Clustering is an unsupervised data mining task which groups objects in the unlabeled dataset based on some proximity measure. Many nature-inspired population-based optimization algorithms have been employed to solve clustering problems. However, few of them lack in balancing exploration and exploitation in global search space in their original form. Differential Evolution (DE) is a nature-inspired population-based global search optimization method which is suitable to explore the solution in global search space. However, it lacks in exploiting the solution. To overcome this deficiency, few literatures incorporate local search algorithms in DE to achieve a good solution in the search space. In this work, we have performed a comparative study to show effectiveness of local search algorithms, such as chaotic local search, Levy flight, and Golden Section Search with DE to balance exploration and exploitation in the search space for clustering problem. We employ an internal validity measure, Sum of Squared Error (SSE), to evaluate the quality of cluster which is based on the compactness of the cluster. We select F-measure and rand index as external validity measures. Extensive results are compared based on six real datasets from UCI machine learning repository.

Keywords Differential evolution · Data clustering · Chaotic local search
Levy flight · Golden section search

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1 Introduction

Clustering groups the objects of an unlabeled dataset [1] based on some similarity measure [3], which plays a significant role in applications in diverse fields. Clustering quality is often assessed by an internal validity measure, which may be based on different features of the clusters, e.g., compactness, isolation, and connectedness. Partitional clustering is one of the most commonly used approaches to solve clustering problems, which directly decomposes datasets into a number of clusters. It can be hard or fuzzy [5, 6].

In the fuzzy partitional clustering, an object may belong to every cluster with some fuzzy membership weight [5]. It is more suited to the datasets having overlapping clusters. Hard partitional clustering groups the objects of datasets into a number of nonoverlapping clusters by assigning an object to only one cluster [6]. It can be mathematically represented as

$$C_p \neq \phi \quad p = 1, \dots, k \quad (1)$$

$$C_p \cap C_q = \phi \quad p, q = 1, \dots, k \text{ and } p \neq q \quad (2)$$

where $C = \{C_1, C_2, \dots, C_k\}$ is a set of k number of non-empty and nonoverlapping clusters.

Many conventional/traditional algorithms for partitional hard clustering, e.g., K-means and k-Medoid, are available to solve the clustering problems as they are simple and easy to implement. However, they suffer from many drawbacks, e.g., the final solution is dependent on the initial solution, or they are easily trapped into a local optimal solution [8].

Specifically for these reasons, nature-inspired optimization methods offer more effectiveness to overcome the deficiency of the conventional clustering methods [4]. They possess several desired key features, e.g., parallel nature and self-organizing behavior [7].

They work on two key aspects, exploration and exploitation to search a good solution in the search space. Exploration is responsible for exploring solutions in the search space, whereas exploitation optimizes the solution locally so as to search a better solution near a good solution. Differential Evolution (DE), a popular global optimization method, provides a good exploration of solutions in the search space. Nevertheless, it lacks in the exploitation of solutions [1, 2, 9, 10].

To balance exploration and exploitation in global search space, few literatures incorporate local search algorithm in DE. In [10], DE is employed to diversify solutions in the search space. However, the local search algorithm, Golden Section Search (GSS), is responsible for refining the solutions in local search space. In addition, few local search algorithms Levy Flight [9] and chaotic local search [1] algorithms are also effectively employed in the literature to solve diverse problems. In this paper, we compare the effectiveness of DE with well-known local search algorithms chaotic local search (CLS), levy flight (LF), and golden section search (GSS) for data clustering. DE assists in exploring solutions in search space, and the local search algorithm

exploits the solutions in the search space to obtain a refined solution. To the best of our knowledge, there has been no attempt in the literature, which reported performance study of DE with these local search algorithms for data clustering in terms of quality of solutions and convergence speed to achieve the final solution. Here, the number of clusters is known a priori in the datasets. We experiment on six real datasets from the UCI machine learning repository to judge the efficacy of these algorithms. F-measure (FM) [6] and Rand Index (RI) [8] are used to judge the accuracy of the obtained clusters. This study will be helpful to the researcher in extending DE to solve diverse real-life problems.

Rest of the paper is organized as follows. Section 2 presents a brief introduction to DE. The proposed method is detailed in Sect. 3. Section 4 presents a comparative study and discussion of the results. Finally, Sect. 5 concludes and remarks upon the possible future research directions.

2 Algorithm Background

In this section, a brief introduction of differential evolution is presented.

2.1 Differential Evolution

Storn and Price [11] propose DE which is a well-known population-based nature-inspired optimization algorithm for multi-model and continuous-valued problems. The basic DE strategy can be expressed as the notation DE/x/y/z, where x represents the target vector to be mutated (a random vector or the best vector), y is the number of difference vectors, and z indicates the crossover scheme [8, 11]. DE encompasses three operators namely mutation, crossover, and selection, which are mentioned as follows:

- **Mutation:** For each individual X_i of current population, a trial vector T_i is created by the mutation operator. In this sense, target vector X_{i1} is mutated with a weighted difference vector of randomly selected two individuals X_{i2} and X_{i3} such $i \neq i_1 \neq i_2 \neq i_3$ as shown in Eq. 3.

$$T_{i,d}(t) = X_{i1,d}(t) + \beta(X_{i2,d}(t) - X_{i3,d}(t)) \quad (3)$$

where d represents the dth dimension of an individual and $\beta \in [0, 1]$ is a mutation-scale factor, which manages the amplification of difference vector; t represents the current generation number.

- **Crossover:** Most frequently used crossovers in DE are binomial crossover and exponential crossover. Here, offspring vector X_i^0 is generated using trial vector T_i and parent vector X_i using the binomial crossover as shown in Eq. 4.

m_{11}	m_{21}		m_{d1}	m_{12}	m_{22}		m_{d2}		m_{1k}	m_{2k}		m_{dk}
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Fig. 1 Solution representation

$$X'_i(t) = T_{id}(t) \text{ if } \text{rand}(0, 1) < CR \text{ or } d = \text{rand } i(1, \dots, D) \\ X_{id}(t) \text{ otherwise} \quad (4)$$

where d represents the d th dimension of a vector and D is total number of dimensions in a vector.

- **Selection:** Apart from selection of individual for the mutation operation to produce the trial vector, selection operation is also performed between offspring vector and parent vector to promote fitter solution to create population for the next generation.

$$X_{id}(t+1) = X'_{id}(t) \text{ if } F(X'_{id}(t)) < f(X_i) \text{ for minimization problem} \\ X_{id}(t) \text{ otherwise} \quad (5)$$

DE attracts attention to researchers mainly because it requires very few parameters to tune and is easy to implement.

3 Designed Clustering Methods

In this section, DE with local search algorithms for data clustering problems is elaborated in detail.

3.1 Solution Representation and Initialization

A centroid-based representation to represent a candidate solution for clustering is followed as discussed in [7]. It is shown in Fig. 1. Here, every candidate solution holds $k \times d$ dimensions, where k denotes the number of clusters and d indicates the number of features in the dataset.

Here, m_{ij} represents the centroid of the j -th cluster and i indicates the i -th feature of the dataset in the solution representation. To commence the initial population for competing algorithms, the dimension of a cluster centroid of each solution is assigned by a random number between the maximum ($x_{\max}$) and minimum ($x_{\min}$) value of that respective feature of the dataset. In Algorithm 1, solution (i, j) represents the j -th feature of i -th cluster centroid; $\text{rand}(0, 1)$ is a uniformly distributed random number in the range $[0, 1]$.

Algorithm 1: Initialization of solutions

```

1: k  number of clusters;
2: d  number of dimensions;
3: for i=1:k do
4:   for j=1:d do
5:     Solution (i; j) = xmin(j) + rand(0; 1) * (xmax(j) - xmin(j));
6:   end for
7: end for

```

3.2 Sum of Squared Error

Internal validity measure represents the quality of a solution. As Sum of Squared Error (SSE) works well with well-separated and compact cluster [5], we select SSE as internal validity measure to evaluate quality of a solution during run. The lower value of SSE indicates the better quality of solution. SSE is expressed in Eq. 6.

$$SSE = \sum_{i=1}^k \sum_{\forall x_0 \in c_i} \|x_0 - m_i\|^2$$

Here, x_0 denotes p -th object of the dataset, m_i denotes centroid of the i -th cluster, and c_i denotes the i -th cluster.

3.3 Procedure for Clustering Methods

To employ designed algorithms for clustering problem, initial population (of size N) is generated. Each solution in the population represents a set of cluster centroids $C = \{C_1, C_2, \dots, C_k\}$, where k is the number of clusters. In this work, DE is selected as global optimization algorithm and DE/rand/1/bin scheme is followed among the several schemes of DE algorithm where DE stands for differential evolution; rand indicates that the target vector is selected randomly; 1 stands for the number of differential vectors selected; and bin stands for binomial crossover. Here, the employment of DE provides diversified solutions in the search space. However, DE exhibits a good exploration of solutions in the search space on the cost of exploitation. Therefore, local search algorithms with DE are incorporated. In this procedure, at the end of each iteration of DE, the obtained best solution is refined by a local search algorithm. Upon incorporating chaotic local search [1] with DE, it is named as DE-CLS.

Similarly, upon incorporating levy flight [9] with DE, it is named as DE-LF. When golden section search [10] is incorporated with DE, it is named as DE-GSS. In each designed algorithm, DE explores the solutions and local search algorithm exploits the solutions to improve solutions in the local search space (neighborhood) of an

obtained solution in the search space to balance exploration and exploitation. The general procedure for these DE-based algorithms is shown.

Algorithm 2: DE for data clustering

```

1: Inputs:
2: Algorithmic parameters;
3:  $n \times d$  data set, where  $n$  is the number of objects in the data sets and  $d$  is the dimension;
4:  $K$  is number of clusters;
5: Steps of algorithm:
6: Initialize the solutions and evaluate their fitness;
7: while (Max_nffe is not reached) do
8:   for For each solution  $x_i$  do
9:     Generate trial vector  $T_i$  by applying mutation operation (refer equation 1.3);
10:    Generate offspring  $x_i$  by binomial crossover (refer equation 1.4);
11:    Evaluate fitness of offspring;
12:    Promote fitter in  $x_i$  and  $x_i$  for next generation.
13:   end for
14:   Select the best solution in the population and employ local search procedure;
15: end while
16: Output: A set of  $K$  clusters of the best solution;

```

4 Experimental Results and Discussion

This section presents experimental setup, description of datasets, clustering results, and comprehensive discussion. The experiments have been performed on a system with core i5 processor and 2 GB RAM in Windows 7 environment using programs written in MATLAB R2012a.

4.1 Parameters Setting

Results of the nature-inspired algorithms are influenced by the number of control parameters. Here, we perform experiments with the set of parametric values (refer Table 1) for DE to perform a comparative study among DE-CLS, DE-LF, and DE-GSS for clustering. Parameter settings of the local search algorithms are set as suggested by the researchers in the respective research papers. Therefore, the number of solutions equals 40. The stopping criterion, i.e., maximum number of fitness function evaluation (Max nffe), is fixed to 10,000 as further processing is merely a computational overhead owing to negligible improvement.

Table 1 Control parameters

Name of parameters	Value
Population size	40
Max nffe	10000
Number of independent runs	25
Scaling factor of DE ()	0.5
Crossover rate of DE (CR)	0.9

Table 2 Datasets

Datasets	Number of clusters (k)	Number of dimensions (d)	Number of objects
Iris	3	4	150
Wine	3	13	178
Zoo	7	16	101
Dermatology	6	34	358
Yeast	10	8	1484
CMC	3	10	1473

4.2 Datasets Descriptions

The datasets are in matrix of size N with real-valued elements which are to be partitioned into k nonoverlapping clusters. Six real datasets from the UCI machine learning repository are selected. A brief summary of these datasets is presented in Table 2.

4.3 Comparison of Results

Performance of these competing algorithms is measured by the fitness function value (FFV) of solutions. The FFV of a solution is the value of SSE which indicates the quality of clusters in terms of compactness. A lower value of the FFV indicates a better solution and a lower value of the standard deviation of the FFV in different runs represents robustness of the algorithm.

Table 3 reports the quality of the solutions of the competing algorithms DE-CLS, DE-LF, and DE-GSS in terms of the best solution—the solution that achieves the lowest FFV—along with the mean and standard deviation (Std dev) of the FFVs obtained in all the independent runs. The convergence speed of these algorithms is shown pictorially in Fig. 2a–f to achieve the best-obtained solution during run.

Table 3 The best, mean, and standard deviation (Std dev) of FFV of final solutions obtained in all independent runs

Datasets		DE-CLS	DE-LF	DE-GSS
Iris	Best	79.22	79.2	79.07
	Mean	82.12	82.85	82.44
	Std dev	2.18	3.38	2.25
Wine	Best	2373962.28	2372750.86	2376829.84
	Mean	2380406.99	2383700.67	2385026.33
	Std dev	4225.23	6398.66	7398.51
Zoo	Best	191	189.48	211.91
	Mean	227.74	220	231.42
	Std dev	21.41	23.72	13.2
Dermatology	Best	22905.22	23076.32	20782.67
	Mean	23784.31	23652.6	22520.23
	Std dev	707.32	440	793.87
Yeast	Best	83.19	84.71	87.62
	Mean	90.7	88.99	92.19
	Std dev	3.64	3.41	2.79
CMC	Best	24376.11	24150.26	24515.31
	Mean	25184.22	25455.98	25347.75
	Std dev	501.02	612.7	713.28

In Table 3, the solutions obtained by the DE-LF outperform other competing algorithm on the datasets Wine, zoo, and CMC. Moreover, its convergence speed is also most promising on these datasets (refer Fig. 2a–f) with respect to other competing algorithms. DE-GSS achieves best value of FFV over other competing algorithms on the datasets, iris, and dermatology. However, DE-CLS produces better quality solutions on only one dataset, yeast with respect to other competing algorithms. In addition, DE-CLS achieves better value of mean and standard deviation (Std. dev) of FFVs in all runs on the datasets Iris, Wine, and CMC with respect to other competing algorithms.

Table 4 presents the F-measure and rand index value of final achieved solution by competing algorithms in all runs. DE-LF has the highest F-measure (FM) and rand index (RI) values over other competing algorithms for the datasets iris, wine, yeast, and CMC; there is only one exception that DE-GSS achieves the highest rand index value for yeast dataset. DE-GSS and DE-CLS report highest values of F-measure and rand index for datasets, zoo and dermatology, respectively.

In Table 3, in case of zoo dataset, though DE-CLS does not achieve better solution (lower FFV value) than the DE-LF, it achieves lower FM and RI value. It means that FFV does not exactly map FM and RI value on these datasets. It is because SSE has no absolute correlation with FM and RI when actual data distribution is not regular. In case of dermatology dataset, based on F-measure and rand index values, all adopted algorithms perform highly unsatisfactory as it is a high-dimensional dataset

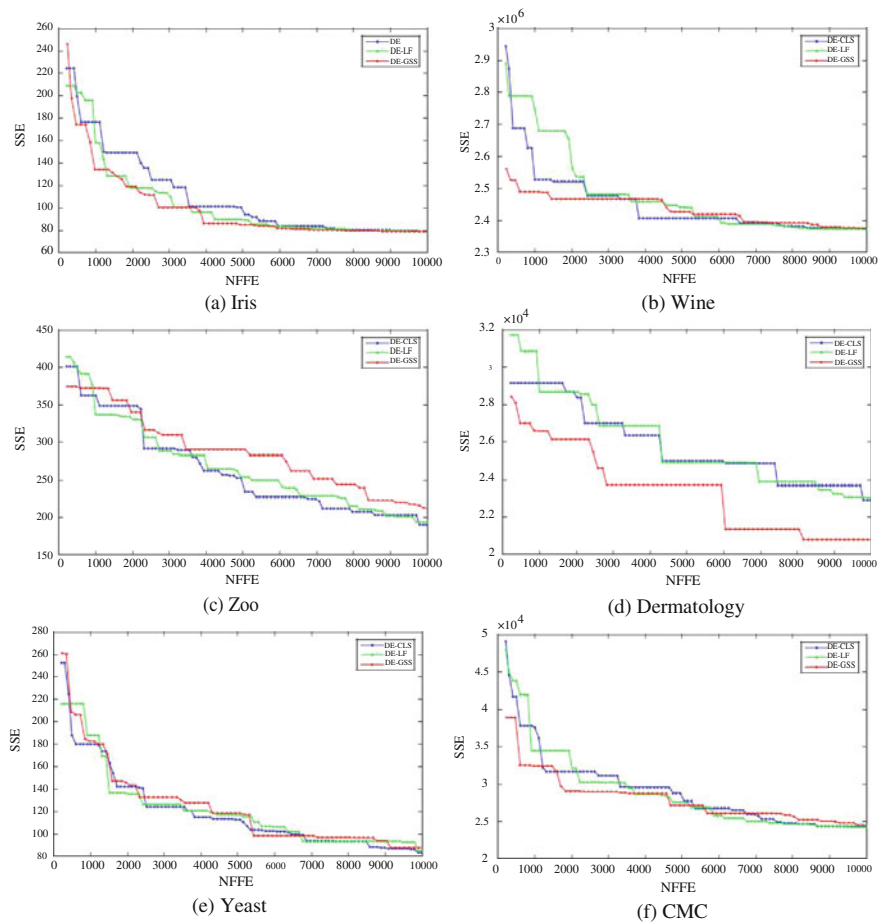


Fig. 2 Convergence speed of algorithm for the datasets

containing 34 dimensions. Overall, DE-LF achieves better value of FFV and highest values of F-measure and rand index on most of the datasets, which indicates that DE-LF balances exploration and exploitation better in global search space. Hence, it is a better alternative to solve clustering problems. However, DE-CLS is more robust as compared to other competing algorithms in terms of achieving FFV of the final solution.

Table 4 The F-measure and rand index of best solution (as per FFV)

	DE-CLS		DE-LF		DE-GSS	
Datasets	F-measure	Rand index	F-measure	Rand index	F-measure	Rand index
Iris	0.8853	0.8737	0.8918	0.8797	0.8853	0.8737
Wine	0.7148	0.7187	0.7148	0.7187	0.7148	0.7187
Zoo	0.7855	0.8865	0.7728	0.8745	0.7414	0.8566
Dermatology	0.2964	0.6919	0.2421	0.6994	0.3099	0.7012
Yeast	0.4403	0.5549	0.4668	0.5737	0.3978	0.6085
CMC	0.4002	0.5578	0.4012	0.5582	0.4002	0.5578

5 Conclusion

Evolutionary algorithms are global search optimization methods to search near-optimal solution within reasonable time. In this paper, a comparative study on DE, an evolutionary algorithm, is performed with well-known local search optimization algorithms, namely chaotic local search (CLS), levy flight (LF), and golden section search (GSS) using internal validity measure SSE for data clustering problems. Effectiveness of these algorithms is compared based on the solution quality and their convergence speed to achieve the final solution in the clustering domain over six real datasets from UCI machine learning repository. External validity measures, F-measure, and rand index are employed to evaluate the quality of final achieved solutions. Results demonstrate that DE-LF outperforms other competing algorithms in most of the datasets. However, in some cases, DE-CLS and DE-GSS provide better results than the DE-LF. Therefore, it is concluded that the DE-LF is a simple and potential algorithm to solve hard partitional clustering.

References

1. Bharti, K.K., Singh, P.K.: Chaotic gradient artificial bee colony for text clustering. *Soft. Comput.* **20**(3), 1113–1126 (2016)
2. Das, S., Mullick, S.S., Suganthan, P.N.: Recent advances in differential evolution—an updated survey. *Swarm Evol. Comput.* **27**, 1–30 (2016)
3. Hruschka, E.R., Campello, R.J.G.B., Freitas, A.A., De Carvalho, A.P.L.F.: A survey of evolutionary algorithms for clustering. *IEEE Trans. Syst. Man Cybern. Part C Appl. Rev.* **39**(2), 133–155 (2009)
4. Rajpurohit, J., Tarun Kumar Sharma, A.A.V.: Glossary of metaheuristic algorithms. *Int. J. Comput. Inf. Syst. Ind. Manag. Appl.* **9**, 181–205 (2017)
5. Jain, A.K., Murty, M.N., Flynn, P.J.: Data clustering: a review. *ACM Comput. Surv. (CSUR)* **31**(3), 264–323 (1999)
6. Prakash, J., Singh, P.: An effective multiobjective approach for hard partitional clustering. *Memetic Comput.* **7**(2), 93–104 (2015)
7. Prakash, J., Singh, P.K.: An effective hybrid method based on de, ga, and k-means for data clustering. In: *Proceedings of the Second International Conference on Soft Computing for Problem Solving (SocProS 2012)*. Springer, pp. 1561–1572, 28–30 Dec, 2012

8. Prakash, J., Singh, P.K.: Evolutionary and swarm intelligence methods for partitional hard clustering. In: 2014 International Conference on Information Technology (ICIT), pp. 264–269. IEEE (2014)
9. Sharma, H., Jadon, S.S., Bansal, J.C., Arya, K.: Levy flight based local search in differential evolution. In: International Conference on Swarm, Evolutionary, and Memetic Computing, pp. 248–259. Springer (2013)
10. Sharma, T.K., Pant, M.: Golden search based artificial bee colony algorithm and its application to solve engineering design problems. In: 2012 Second International Conference on Advanced Computing & Communication Technologies (ACCT), pp. 156–160. IEEE (2012)
11. Storn, R., Price, K.: Differential evolution—a simple and efficient heuristic for global optimization over continuous spaces. *J. Global Optim.* **11**(4), 341–359 (1997)